

An exploration of Interactive Technology in bone strengthening exercise for sedentary office workers

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Abstract

A lack of physical activity due to non-WHO regulated education and desk jobs is a large problem, affecting most, if not all, white-collar lines of work. The resulting lack of activity creates a high percentage of people who do not practice any sport, putting themselves at a severe health risk. In this report, we seek to show the process of developing a system to help counteract this issue, and how our vision evolved over time. Initially, we believed that we would find an explanation by studying people who quit the gym, predicting it would be tied to boredom and a small amount of socialising. However, upon researching the topic and conducting interviews with such people, we came to the realisation that this hypothesis was erroneous. Consequently, this led us to study autonomy issues, and better understand what drives a person to be active, with the aim of using this knowledge to design a device which let people with sedentary lifestyles practice a form of sport which was easily accessible. In order to achieve this, we opted for the creation of a game which used physical equipment as a controller.

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1. Introduction

Continuous physical activity has been shown to boost not only physiological well-being, but also mental well-being. Going to the gym is a popular choice for people trying to be physically active and remain in good health. However, while gymnasiums might provide all the necessary resources to work out effectively, many who start going, eventually stop further down the line. In certain cases, upwards of 89% of those who signed up for the gym eventually dropped out within the year (Oliveira et al., 2021). The reasons range from not having enough time, to being overwhelmed by the equipment (Radhakrishnan et al., 2020). In summation, due to this high abandonment rate, not enough people are physically active on a regular basis. This is seen especially amongst office workers, who suffer from reduced levels of health and productivity due to their sedentary work obligations (Gonzales et al., 2022). Due to the high inertia in maintaining regular exercise, many office workers do not end up practicing any kind of adequate physical activity. They will instead spend upwards of 10 hours a day sitting down (Biernat et al., 2010).

This is a serious health risk, since around 3.3 million people die annually as a result of insufficient regular physical activity (Jansson et al., 2019), through various types of heart disease, high blood pressure hypercholesterolemia (Altavilla, 2016). Furthermore, this shortfall has a significant impact on bone strength, which can then lead to diseases such as osteoporosis, which entail high morbidity and increased mortality (Ammann & Rizzoli, 2003). To combat this inactivity, outdoor gyms have been installed in increasing numbers in town parks as government-funded installations. This is to attempt to bring sports equipment closer to the people (Fernández-Rodríguez et al., 2020). While these installations might seem like a good solution, they are mostly used by the older generations (Chow et al., 2017), and not by office workers, who remain unmotivated to carry out continuous repetitive exercise regimes, in spite of more open access to equipment.

In response to this widespread difficulty to maintain a sufficient quantity of physical activity, our team has taken an interest in tackling the issue of the exercise deficit seen in office spaces, particularly focusing on the field of practices that lead to the maintenance or increase of bone density. To overcome the lack of continued motivation seen in sedentary employees, we have endeavoured to make a participative game to encourage them to persist in their physical activity. Our team decided to use interactive technology to increase the autonomy of white-collared workers. The goal of the research was to produce a device, or activity, which was both easy to engage with, and intractable in a way that remained brief enough to not overexert those

who used it. The intention was not to entirely replace existing sports and activities, but instead to provide office workers with an easy way to carry out short bursts of physical activity during a workday, without needing to dedicate too much time and effort. Thus, we posed the following question:

What is the correlation between the accessibility of providing gym equipment in office spaces and the average level physical activity seen in the population of office workers?

Through research and discussion, our team has developed a device which can be placed in office breakrooms, and act as a point of leisure for the workers which are passing by. It takes the shape of a simplistic game, which is controlled by a rubber band which has sensing features capable of detecting the direction and magnitude of the rubber band. Using this controller, players need to control and aim an imaginary bow and hit targets by holding the rubber band in a stressed position for a certain amount of time. The goal is to make players exert themselves using a resistive exercise, while not being excessively tiring to use for a break period, since the office workers then need to continue working after these breaks.

2. Context analysis and concept

2.1. Research Questions

First phase research questions

The sub-research questions are:

- What are the reasons people start exercising?
- What are the main reasons for people to stop their physical exercise regime?
- What keeps people motivated to continue physical activity on a long-term basis?

The phase one research question is:

What are the main reasons for people to stop their physical exercise regime?

Second phase research questions

1. Can interactive technology, in conjunction with readily accessible physical exercise equipment, be utilized to make exercises more interesting in order to sustain physical activity amongst office workers who are insufficiently active?
2. What factors contribute to office workers not engaging in enough bone-strengthening exercise, according to WHO guidelines?

2.2. Approach

We began our research by attempting to answer the first three sub-research questions, which delineate our main focus area, and must be addressed through literature research. This method is the most convenient, as the questions we ask are not novel, meaning there are many existing studies to help further our understanding of the subject. Furthermore, these studies also provided valuable quantitative data, which helped us measure the scale of certain variables.

In order to find these studies, we mainly used Google Scholar and Scopus, which are large databases of scientific studies, including the realm of Human Computer Interaction, and made a set of rules to best optimize our findings. We began with a couple of key search terms, such as gym AND motivation, or gym AND dropout, which produced a large number of studies, which would take excessively long to read through to find the appropriate research. For this reason, the search was then narrowed by adding more key terms, progressively nearing the most suited papers for our research, until there was a reasonable quantity of studies to read through comprehensively. In addition to this search method, we picked a set of criteria which would help narrow the set of results, such as relevance to our topic, or the recency of the paper, limiting the results to papers published between 2004 and 2024. Another important criteria was the legitimacy of the paper, which we determined by observing how many other papers cited it, and the significance of the study group sizes. Larger subject pools provide a more accurate dataset therefore validating the findings of the research project in question. The scientific rigor utilized to select appropriate papers related to our project enabling us to answer our leading sub-research questions with a high level of confidence. We excluded papers that used less than 17 subjects for their research, since this number of subjects is the threshold at which results become saturated (Francis et al., 2009).

The selection process consisted of reading through the already filtered selection of results, amounting on average to fifteen different papers discussing similar questions, which then got reduced to one or two relevant papers after applying our criteria. In the cases where the two remaining papers were similar in terms of legitimacy and relevance, we picked the most recent one.

Based on the results of other studies, we concluded that one of the main reasons for non-continuation of physical activity sessions was lack of time during a regular work day to fit all obligations in. Due to this fact, we decided to change our focus from the specific idea of finding interactive technology to stop individuals from dropping out of the gym, to a more general

attempt to promote physical activity of office workers by using interactive technology in extremely accessible areas, such as office hallways. We constructed our research question for the project's first phase:

“How significant is the link between regular physical activity and the level of health of workers with sedentary jobs?”

For this research question, we decided to conduct interviews both with people who had stopped exercising, and who still continue to do regular physical activity. These interviews were elicited to provide us with qualitative data, which was crucial for the design of a product that could solve the user's issue with lack of autonomy and decreasing bone density.

To find candidate users for the interviews, we reached out to individuals who currently go to the gym or have stopped recently. Before interviews, we asked them to sign a paper consent form.

After conducting and analyzing the interviews, we asked our first research question: “How effective is accessibility of equipment in sustaining physical activity?”

In order to answer this question, additional research into the topic was needed, which led to a comprehensive search for papers which included the words “outdoor exercise facilities usage” on Google Scholar, using the aforementioned criteria to determine the best papers.

With the insights from the interviews as well as the new knowledge from additional literature research, we changed the direction of our project from making generalized exercising more interesting to providing people with an opportunity to exercise to maintain or increase their bone density without the need to invest much time or resources. With that, the second research question of phase two was formulated: “What factors contribute to office workers not engaging in enough bone strengthening exercise, according to WHO guidelines?”

Due to some variables related to this topic being unknown, we came to the decision that interviews needed to be conducted on the target users to gather the required data. Through this, we could generate more valuable data regarding the user group, their needs and the specifics of their exercise routines. Further details about the interviews will be discussed in section 2.4.

2.3. Context Analysis: Literature and Related Work

Physical exercise is necessary for good health and wellbeing, something that Elmagd (2016) stresses in particular, going to the extent of saying that exercising daily would reap a considerable number of benefits. These benefits range from reducing anxiety, to decreasing the risk of cancer, showing the importance of physical activity. However, there are many cases in which people do not practice any kind of sport. Team sports have been shown to increase sustained physical activity (Pedersen et al., 2017), however for those who decide to exercise alone in environments, such as gyms, it can be difficult to remain consistent with routines when the program consists of a repetition of exercises with little novelty during each session. Research has shown that only 5% of members renew their membership after a year (Sperandei et al., 2016). Looking at a more general scope, beyond the gym, Biernat et al. (2010) reveal in the worst cases, up to 70% of employees in a certain line of work did not practice sufficient amounts of physical activity, exhibiting instead a sedentary lifestyle, devoid of physical exertion. This occurred regardless of the quantity of locales available that were dedicated to physical activity and was shown to remain a constant throughout the years, becoming even more of a problem in more recent years. The most common reasoning among office workers for their lack of physical activity was due to a lack of time as an external barrier, and a lack of energy as an internal barrier (Edmunds et al., 2013), showing that office workers are not positively responsive to most forms of physical exercise. The combination of lack of time and diminished energy often results in the person choosing to overlook this facet of wellbeing. This creates a serious health risk, not only for physical health but idem for mental health, as overworking has been shown to also lead to mental exhaustion, which makes the office workers become significantly less productive (Gonzales et al., 2022).

This becomes problematic, as the incurring dangers from a lack of physical exercise are significant, such as low cardiorespiratory fitness, which at one point caused 16% of all deaths in the USA (Blair, 2009), outnumbering even smoking and hypertension. On top of that, this lack of physical activity has an impact on bone strength, which can then lead to diseases like osteoporosis, which entail high morbidity and increased mortality (Ammann & Rizzoli, 2003). In addition to conditions directly caused by the lack of such physical activity, it also worsens the severity of already existing conditions (Bull et al., 2004), putting people from any age range at risk. The gravity of the consequences makes it imperative that the physical wellbeing of people is prioritised for future health products. Inversely, as Elmagd previously discussed, the benefits

of doing regular training are considerable, effectively reducing the risks of death from any cause by over 50% (Warburton et al., 2006).

Given these findings, it is worth reflecting upon the reasons people might have to not do sport, despite the positive effects. For this, we take the example of doing repetitive exercises in environments such as the gym, which, along with jogging, remains a common method for people to retain their fitness.

Self Determination Theory (SDT)

As explained by Ryan et al. (2009), SDT is composed of three basic human needs which fuel people's motivation: autonomy, competence, and relatedness. In order to sustain their gym attendance, people need to improve at least one of these basic human needs, so that their time feels well spent. In fact, Caudwell & Keatley (2016) elaborate on this, stating that an individual's explicit autonomous motivation was the greatest predictor of the continued physical activity with repetitive exercise, alongside attitude. After noticing some statistical trends, Crossley (2006) began exploring in further detail the many reasons people had for joining the gym, additionally investigating what were the key motivators which kept these same people engaged in repetitive exercise sessions for extended periods of time. The results of this search revealed that a common line of reasoning was the desire to be healthier, specifically in terms of appearance. While insecurity about body appearance might be an initial motivator to start repetitive exercise, Crossley found that this dominating reason was eventually replaced by an enjoyment for the gym in the long run, followed by a want for social interaction. Many gym goers reported that their motivation increased from the effects that it had on their self-esteem and their "physical self," as Crossley called it. These results are then confirmed by Söderström (2023) which found that gym goers are most often motivated by their goals and the pleasure they get from going, insinuating that making exercise a more pleasurable experience could help some individual to retain motivation in their own activities.

On multiple occasions, attempts to promote physical activity and movement which using SDT appeared, like the fitness app designed by Gui et al. (2017) which intended to help motivate people by using their need for relatedness. Acting as a fitness social media, it let users share the data from their activities with others, connecting people and improving their desire to be consistent. Among these inventions that attempt to increase the overall amount of physical activity in a population, the most widespread development which has risen significantly over the years has been outdoor gyms (Cranney et al., 2016). These installations have been placed in

accessible areas of towns and cities with the goal of substantially reducing the travel time needed and lowering the bar of entry for gym-styled physical activity. These address the autonomy part of intrinsic motivation, eliminating many of the mental barriers that most other activities have, by being typically close and easily accessible (Fernández-Rodríguez et al., 2020). However, such outdoor gym designs suffer from a few limitations according to Batesa et al. (2013). The paper points out that the equipment can easily get wet, and that there is insufficient sitting space. While these issues can be remedied, they affect how people view these areas as places to participate in serious physical exercise. However, a more abstract issue that was recorded is confidence, defined by the user's fear of public embarrassment if they do not know how to use a certain machine. Batesa et al. suggest that the equipment should include instructions for usage, but this would only lessen the issue and not mitigate it.

In a project developed by Ren et al (2018), office workers were encouraged to increase their physical activity by creating cooperative fitness tracking. This attempt used the relatedness part of the SDT to motivate office workers to move more. With this tracking, the researchers hypothesized that cooperative tracking would improve the awareness of being physically active. Improving the mindfulness of office workers in relation to their physical activity has been used in other solutions, such as visualizing computer-based activity to make office workers aware of their sedentary behaviour (Brombacher et al., 2020).

Our aim was to design a system which would be extremely accessible, and intuitively easy to use. The end goal is to improve the autonomy of users, therefore leading them to increase the time spent on sporting activities via the operation of an installation that makes use of interactive play, as explained by Delden et al. (2018). To corroborate this idea, research has shown that the medium of video games has proven to be effective at increasing people's motivation, as well as their wellbeing (Ryan et al., The Motivational Pull of Video Games: A Self-Determination Theory Approach, 2006). In their paper on improving gamification, van Roy & Zaman (2017) discuss the main flaws it has in its current state, and imposes key heuristics based on the three pillars of the SDT: autonomy, competence and relatedness. Our focus is on autonomy, which signifies that our goal should be to make the participants feel like they are using the device out of their own volition, for the amount of time they feel works best for them.

2.4. Context Analysis: Interviews and Observations

To gain more data on our user group, we performed a round of interviews with potential users. At this stage our user group was students, so it is important to keep in mind that they are not our

end users, office workers are. However, the information gained about gym-goers and dropouts was still useful. To better understand the internal thoughts and motivations of our user group, we asked questions regarding their goals, problems, and their reasons behind why they exercise, or inversely why they do not. To produce finer details concerning the subject's motivations to be physically active, we asked specific questions about their exercise regime, and their success in maintaining a repetitive exercise program.

Method and participants

Participants for the interviews were gathered through friends and family of the researchers who met the criteria. A total of ($n = 13$) participants were interviewed. Ten of the interviewees were students at the University of Twente, aged 19-26, and the other three were parents aged 54-56. Six of the participants were male while the other seven were female.

In the interviews conducted, a semi-structured approach was used. While the interviewer did go through a set list of questions, there was also room for additional questions if more context was needed. The list of main questions used in the interviews is included in Appendix 2.2. At the time these interviews were conducted, the main user groups for the project were gym goers and gym dropouts, hence why the chosen participants reflected this demographic, however this has since changed. Despite this, the results must be included. At the time of the interviews, 7 participants were attending gyms where they did repetitive exercises, and 6 participants had stopped exercising. 6 participants were interviewed online, while 7 participants were interviewed in person. The interviews' audio was transcribed using the online website Turboscribe, a platform designed for transcription and documentation.

Analysis

With the transcripts of the interviews, we reviewed the responses of all the participants, and structured them into a cohesive summary, which grouped the points of interest together into their respective themes. Some questions had a set number of potential responses, in which case the choice frequency of a certain response was tallied, to see if one appeared more often than others. The summarized results of the interviews can be found under Appendix 2.4.

The results reveal that in many cases, dropping out from the gym was not an immediate or deliberate decision, as reported by 5 out of the 6 dropout participants. These participants described their eventual abandonment of the gym as a slow process in which they went progressively less frequently, with one participant taking upwards of 6 months.

The leading causes interviewees gave were the following:

- Time-schedule issues
- Travelling to the gym
- Waiting for machines

Based on the described pain points, we were able to ascertain that one of the major contributing factors to routinely carrying out any kind of physical activity revolved around the concept of autonomy. The perceived benefits of going to the gym were outweighed by other issues that the participants in the interviews had with a component of the experience. From the interviews, we determined that 83% of the people who stopped practicing a regular physical activity had reasons which could be connected to a lack of autonomy, such as travel times and schedule issues. However, despite these issues, a majority of these people reported enjoying their physical activity when they still did it, with only two which did not or weren't sure out of the six. Of these four participants, three claimed to enjoy the visible progress they were making over time with these exercises. The results of the interviews did not match our predictions which we made based on the literature research. Our expectation was that dropouts would dislike going to the gym, with the issue not being related to autonomy. Therefore, due to these findings, the core focus of the research was shifted towards the concept of autonomy, with a particular interest in how the travel time to dedicated exercise locations affects a person's initiative.

Discussion

While the findings from the interviews were relevant to our research and helped steer our goals, the results might have been more accurate with a different methodology and a larger sample size. When reviewing the interviews, only one researcher looked at one given interview transcript, which might have led to researcher bias, potentially altering findings or missing valuable data points. Furthermore, a summary has the potential to miss some of the finer details of an interview.

An alternative method which was considered for this task was a collaborative coding session of all the transcripts, creating a hierarchy of codes and categories based on the responses of the interviewees. Despite taking more time and organization, this method would prove itself more reliable for producing insightful and detailed findings.

The interviews we conducted were flawed and had room for improvement. Firstly, the researchers were novices in conducting interviews, therefore may have not had the experience

needed to carry out the interviews with a more in-depth methodology. A more experienced researcher in the field of collecting data from interview subjects may have been able to ask more insightful follow-up questions in order to gather a more comprehensive range of information. Secondly, questions were asked about past experiences of the participants, such as why they started going to the gym. With reflection, it would have been more insightful to have long-term contact with people starting physical activity for the first time and asking them questions throughout their exercise journey. Time constraints made this infeasible for this research.

2.5. Description of Target User Group and Domain

Based on our observations, we can best define our target user group as white-collar office workers who do not actively have an exercise program. Specifically, the office workers who keep sitting during their breaks. The ones that do not feel the need to go out for a walk or to do any other kind of physical activity, due to a lack of commitment and autonomy. As previously discussed, an absence of physical activity is detrimental to a person's physiological and cognitive capacities. The particularity with this group of users is that due to the constraints of their daily working lives, they have a tendency towards sedentary behaviour over the course of a day, both during work and outside of work (Clemes et al., 2014). Such desk jobs require workers to spend upwards of 75% of their working hours in a seated position, often leading to neck, back and shoulder problems, and exhaustion, as well as more serious health risks such as hypertension or hyperlipidaemia (Daneshmandi et al., 2017).

The technology domain selected for this project was game controllers, as we wanted to present the notion of a casual ambiance for the potential users of the device, while making use of the reported motivation increases from a gamified system.

Earlier research on the effect of inactivity on the general health of sedentary workers has led to product developments that have successfully helped users retain their commitment to physical activity by using gamification. An example of the successful implementation of gamification was observed in the creation of a social application in which users shared their fitness data with other members. As a result, the users experience an increased amount of relatedness between users, which in turn was shown to augment the amount of exercise the active users of the app accomplished (Gui et al., 2017).

2.6. Product Concept and Global Requirements

The first step in the development of the device concept needed to address the issue of office worker inactivity was the creation of a mind map by brainstorming a wide search of existing technologies and user domains that were related to physical activity. The resulting extensive list of interactive technologies already in play allowed our team to delve deeper into the relationship between each technology and user domain. Through discussion, the list was reduced to six technologies and six user domains, which we perceived to have the most potential in the creation of a prototype for an interactive technology product that met the criteria of the study. These technologies and user domains were then placed into a 6x6 matrix which was set up to combine every technology with every user domain. In these spots, we ideated on possible applications which would involve both elements of the intersection, until we had a selection of ideas to pick from.

After studying the matrix, we decided to focus our final user domain on sedentary office workers.

The technology domain we decided to focus on was game controllers. This was chosen because of the nature of the project, as well as the associated playfulness which comes with such a technology.

Using the resulting intersection of domains, we carried out the procedure of a '50 ideas brainstorm'. From the subsequent generation of ideas, our team was then able to discuss and opt for the best three ideas. As the final step in the process, one last '50 ideas brainstorm' was performed to create a comprehensive list of details for these individual ideas.

The final concept of this phase was "Gym party mode". It would be an application that would help people in the gym find spotters, initialize friendly competitions and a way to track their progress. This aligns with the research outcomes which say that people are motivated by social interactions in gyms (Crossley, 2006).

Based on the literature research and related work, we also made a first version of the global requirements:

- The product must intrinsically motivate the user to keep working out.
- The product should not be invasive, and respect the user's privacy.
- The product should not obstruct the workout of the user

From the outcome of the interviews carried out, we were able to ascertain that motivation was not cited as the main issue for individuals attempting to maintain physical activity, but the predominant challenge was autonomy, especially in the area of dedicated time needed to perform the activity. The interviewees specifically mentioned: waiting times, boredom with repetitive exercises, traveling to the gym and finding time to fit their workout in their daily schedule. Based on these findings we altered the concept to aid the autonomy of people desiring to work out but lacking time during their daily obligations. Outside gym facilities do already provide an opportunity to exercise with a lower time requirement, however, users usually view it as a fun activity, rather than a serious exercising facility (Cranney et al., 2016). We decided to use this data as our inspiration and reinitiated the process of the '50 ideas brainstorm'.

The final concept we chose was an arcade-like game system that has a band as a controller that could be deployed in the lounge areas of office spaces, to provide an opportunity to quickly exercise in between daily tasks. The band was chosen to be the controller that would be used for this project, providing a unique gaming experience.

Looking at the related work, there are systems that might be bone strengthening. It was also found there are systems accessible to office workers. Our team decided to work on a system that does both, offering an accessible way to do resistive exercise. Since the product will be used in the context of an

office space, often being used in one's break, the system has to not take up too much space and not require a large time investment to use. Considering this, a list of global requirements was made.

The global requirements for the system were:

- The system must fit in an area of 1 square meter.
- The system must provide an experience of 2-15 minutes.
- The system must be intuitive to use.
- The system must be safe to use.
- The system's components could be modular and modifiable.
- The system must provide an opportunity for light to moderate resistance exercise.
- The system must be accessible to office workers.

3. Design and Evaluation of Lo-fi Prototype(s) and Revised Product Concept

3.1. Lo-Fi Prototypes

Two prototypes were made for this device, covering the key design features identified through the interview process as crucial to test for. Given the nature of the concept, the first element to test was the type of exercise that the users would need to do to operate the device. The goal was to determine which movement the majority of participants would find most comfortable and enjoyable when doing repeatedly, while also being safe to perform. For the movement, functional exercises assisted with an elastic resistance band were chosen, as these are motions that help train the muscles for daily mobility. An elastic resistance band was chosen as it has been shown that exercising with an elastic band helps increase the bone mineral density (Li et al., 2023), and also helps with increasing muscle strength (Vafaeenasab et al., 2019). The advantages of utilizing an elastic band were twofold. Firstly, the band made for an easier interface as a controller compared to dumbbells, and secondly, as the resistance increased with the distance pulled, it simplified access of the equipment for users of different strength levels.

The first question that needed to be addressed was:

- Does the movement performed by the participants increase enjoyment of the activity proposed?

The movement needed to also feel rewarding to the participants involved. To ascertain which motion fitted this criteria, different movements were explored to discern the most popular one to be implemented in the design of the product.

The full research question that accompanied this prototype session was:

- Can an interactive technology device be developed to take advantage of functional exercises, assisted by a resistance band, that people find pleasing in order to motivate them to continue sessions of physical activity on a prolonged basis?

The targeted user experience for this prototype focused on asking the user to experience different movements carried out with some resistance. The execution of these movements came with the requirement that the user feel some sense of accomplishment in performing them due

to the perceived difficulty of the movement involved. The responses to the trialed movements were integral in determining the ideal movement to be applied to the prototype designs.

The initial prototype set up to test the movements was a combination of a simple resistance band attached to a pole at varying heights, and a frisbee. The movements evaluated were pulling the band from top to chest, pulling the band from middle level to chest, pulling the band from bottom to chest, pushing the band from middle level away from the chest, and lastly, rotating the frisbee. To achieve these variations, the resistance band was mounted on a pole at its base, middle level, and then above the heads of the users. The frisbee was held by one of the researchers to emulate resistance while the user tried to turn it. In Figure 1, the scenarios are shown as sketches.

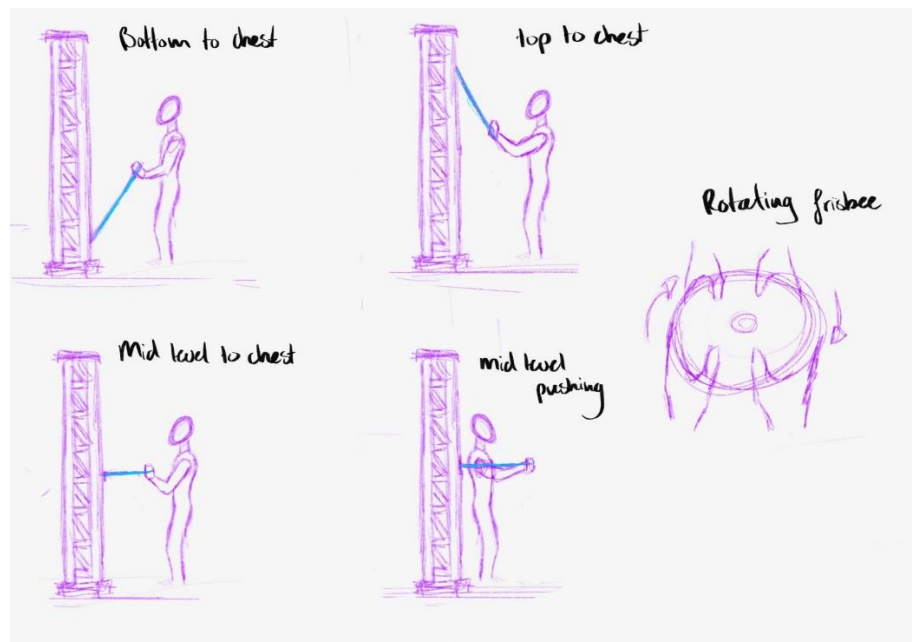


Figure 1 Scenario sketches.

Eight students were selected as participants for this prototype. They were randomly selected on campus. Their background, age, gender and country of birth were inconsequential for this test. The participants were students because at this stage of the research the target user group were students. All the participants agreed to the consent forms.

The evaluation was executed in the SmartXP. The resistance band was placed on a pole and changed in height for each individual activity in the presence of the participants. One of the team members would ask a random student to participate in the testing. Once they agreed, they were admitted into the testing area, with the first obligatory action being to sign the consent form

before participating. After this, the participant was conducted to the resistance band. Before starting the activity, the different movements to be evaluated were explained and demonstrated. Once the participants trialled each motion, the team member changed the height of the band and demonstrated the movement the participant needed to perform. After the activities with the resistance band were completed, the researcher demonstrated the movement to be investigated with the use of a frisbee. The participant was asked to attempt to turn the frisbee in a designated direction while it was being held by the researcher. At the end of the trials, the participant was asked to rank the movements from 1 to 5, depending on which movement felt most comfortable and rewarding, with 1 being the optimal activity.

The results of this evaluation were written down and later put in a table. This table can be seen in figure 2.

FREQUENCY OF GIVEN RANKINGS FOR EACH TYPE OF EXERCISE

PLACE	<i>Pulling up from the ground</i>	<i>Pulling from chest height</i>	<i>Pulling down from eye level</i>	<i>Pushing from chest height</i>	<i>Rotating</i>
1 (100 POINTS)	4	2	2		
2 (80 POINTS)		4	4		
3 (60 POINTS)	2	1	2	2	1
4 (40 POINTS)	2	1		1	4
5 (20 POINTS)				5	3
POINTS:	600	620	640	260	280

Table 1 prototype 1 results

The table above shows how many times a certain movement obtained a certain place in the ranking, e.g. pulling up from the ground was placed first four times. From the table, we were able to deduce that pushing from the chest, and rotating, were the least favourite motions. therefore these two movements were excluded. To determine which of the remaining three movements was the optimal choice for the system we created a measuring system of points and assigned them to certain placings. The points were then added up to calculate which of the active motions were the most favoured by the participants. It was seen from the final results that pulling down from eye level had the highest total. The movements of pulling up from the ground, from chest height, and from eye level had similar results, therefore demonstrating that the capability to change the height of the elastic band was an important feature in subsequent editions of the prototype.

After determining how the users preferred to interact with the system, the next step was to develop the visual and interactive aspects of the product. For the following prototype, different installation designs using a paper prototype were tested. The research question to be answered in this part of the process was: what type of design would people prefer to interact with?

The targeted user experience was to encourage the participants to feel attracted and intrigued by the aesthetics of the product. To evaluate this, we conducted a simple test consisting of three sketches made of different object designs that could potentially be placed in a hallway. The participants were all students randomly selected on campus, which was the target user group at the time of testing. The background, age and gender of the subjects participating in the study was incidental for the selection. These participants were introduced to an imaginary scenario in which they were asked to imagine inadvertently coming across the product in a hallway on campus. They were instructed that their task would be to quickly choose one design of three possible installations that they would be more likely to approach. The three designs can be seen in Figure 3.

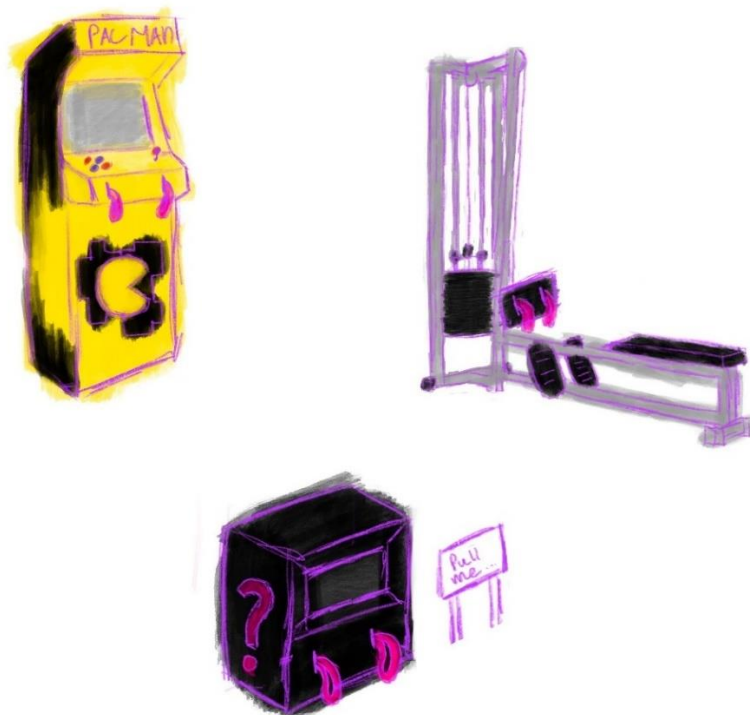


Figure 2 sketches that were shown in lo-fi test 2

In total, seven students were asked to participate in the test, with the results recorded in a word document. The data consisted of how many students chose each object as the most approachable, and some key words why this was the case. The outcome of the test was that 4 students chose to first walk towards the black box, 3 students chose the arcade machine, and no students chose the object designed to adhere to the popular view of gym equipment. The main reason cited for students choosing the black box was the curiosity they felt when seeing that the box looked mysterious due to the question mark and the sign next to it. For the arcade machine, students chose it predominantly due to its bright yellow color, and its familiar gaming design.

From this lo-fi prototype we determined the optimal aesthetics of our product. We decided to construct a black box with a question mark on it, accompanied by a sign with 'pull me' written on it. Using this design, we hypothesized that users would willingly approach the product.

For this project, the design rationale for this device was able to be summarized in a few key points. Given the nature of the device, we wanted it to be both easily accessible and easy to use, to ensure that it catered to as many people as possible, and required the lowest learning arc to be able to begin using it. Furthermore, we wanted to make it useful to strengthen bone density, while being able to have an effect on physical activity in the space of time of an average work pause. However, the key goal for this device was its entertainment and engagement value, reflected in the participant's desire to return to the installation.

3.2. Revised Product Concept and Requirements

Based on the results of the two lo-fi prototyping experiments, we refined our product concept to be a system that can be placed in offices to provide an opportunity to do simple pulling exercises in the time frame of a break.

With the updated concept, we also formulated additional global requirements:

- The user must be able to control the system using the resistance band to perform the pulling exercise. The first lo-fi prototype test shows that the pulling motion was preferred over the pushing or rotating ones.
- The device should be designed in a way that is intriguing to passers-by. The second lo-fi prototype concluded that a design which sparks curiosity and wonder was the most interesting one to approach.

- The system could provide users with the option of adjusting the height of the pulling mechanism. From the first Lofi prototype test the preferred pulling motions were ranked close to each other which indicated that there might be demand for adjustable height settings.

4. Design of Final Hi-Fi Prototype

The final hi-fi prototype consisted of a black wooden box with a question mark painted on its side, in addition to an invitation to interact with it painted on the front. The prototype was constructed to fit on a table inside a breakroom, or hallway of an office building. A hand grip and an elastic band were fixed to the box. The elastic band served as the game controller for the game contained inside the box. The game developed was based on the concept of movements used in archery in which the player used the elastic band to aim and shoot arrows at different targets. A pressure sensor on the band ensured that the player stretched the band enough to engage their muscles in the action.



Figure 3 Final Hi-Fi prototype

Firstly, the global requirements from chapter 2 were considered to make sure that the final created prototype fitted within a designated area of 1 square meter, in order to be easily installed in a breakroom or hallway in an office building. The black box was constructed with the dimensions of 66 x 46 x 24 cm. This requirement was important to ensure that our target users would have simple access to the device, and employers would be able to easily accommodate the system in their buildings. Another important global requirement that was implemented was that the system must take between two to fifteen minutes to use. The final prototype that was built took the users an average of two minutes to perform the activity, showing that the user can

autonomously decide how long they wish to interact with the device. The findings from initial interviews pointed clearly to the conclusion that gyms are often not an option for office workers, due to autonomy related issues, such as time constraints. Hence, our system was designed to lower the threshold of starting an exercise. We achieved this threshold lowering by guaranteeing that the system did not take too long to use, and that the effort required to perform the action asked for was not too difficult for the average person. The final product had multiple levels for the user to finish, with the option of completing them more than once if desired, at the discretion of the individual.

According to the global requirements, the system had to also be intuitive to use. This was important since the product needed to be actionable with minimal instructions in its use. After carrying out experiments on preferred movements on test subjects, an archery game was chosen to be the most intuitive game for a resistance band. When operating an arrow, a strand is pulled back by hand. To mimic the same type of movement, it was necessary that the resistance band should also be pulled back with a hand. The aim of the system to be as intuitive as possible was realized as the device worked like a real archery bow. If the band was pulled back to the appropriate tightness and direction, the arrow was set to shoot and hit the targets on the screen. To help the user perform the action successfully, the box also came equipped with a hand grip to simulate the real use of a bow in which the other hand is used to hold the bow in place. However, the hand grip was not only for intuitive usage, but also installed to make the prototype safer, which was another requirement decided on in the initial process. To obtain market-readiness of the product, the legal requirement for safety had to be fulfilled. Extra measurements were taken to ensure that the box would stay rigid on a table, to avoid the possible scenario of the box falling to the ground and harming the user. To add additional safety restraints, we used two straps that were connected to the box that could be also connected to the table that it was placed on. Additionally, the box needed to have a sturdy frame so that people could pull on the resistive band without breaking the device. A sturdy frame also helped with strapping the box down to a table. To make the frame strong we screwed thick wooden beams together using metal angle connections. At this juncture, the only danger our product still had was that users could strain or pull a muscle.

Another global requirement that was important for this final prototype was that it must provide an opportunity for light to moderate resistive exercise. This was achieved by using the resistance band that was also used for conducting the first lo-fi prototype. Implementing this was important since the focus for this product was also to increase bone strength of office workers. The

exercise available had to be at a light to moderate level to keep the threshold for office workers low and their autonomy high. This was done by using a medium resistance band. On top of that, the office workers had the possibility to independently decide the length of time they dedicated to the exercise. To keep the autonomy high, the system needed to also be accessible to the office workers. This was done by creating a reasonably small product that could be easily put in an office space. Preferably, it was designed to be put in a breakroom or next to a coffee machine. The accessibility meant that office workers did not have to consciously decide to use the system; instead encountering the system when taking a break or getting coffee.

There was also a “could” global requirement that was not implemented in the final Hi-Fi prototype, indicating that the system's components must be modular and modifiable. This requirement would still be an advantageous addition to the system to make it accessible for more people. As an example of a useful feature, the strength of the band could be adjusted to adapt to the different strength levels of each individual user.

On top of the global requirements, we also considered the additional requirements that were formulated in chapter 3, after the lo-fi prototypes. The first additional requirement that was formulated after conducting the first lo-fi prototype was that the user should be able to do pulling exercises while using the system. In addition, we realized that people preferred pulling exercises over pushing and turning. The prototype was conducted using a resistive training band, since our research suggested that it could improve bone density. In light of this finding, we decided to also use this band in our final prototype. With the band the user was encouraged to do pulling exercises. Another requirement based on the first prototype was that the system could provide the user with the option to choose the height of the pulling mechanism. This was another “could” requirement that has not yet been implemented in our Hi-Fi prototype due to time constraints. Having more options for height could make the prototype more inclusive for users.

The system should have an alluring aspect to intrigue and attract the user, which is a requirement that came out of the second lo-fi prototype session. From this session, it became clear that most of the individuals tested would first walk towards an object that was black with a question mark and sign next to it. The unknown quantity and use of the black box woke a sense of curiosity in the potential users, enticing them to approach the device. Some participants commented that the ‘pull me’ sign in particular convinced them to walk towards the box and pull on the band. Therefore, the original idea of creating a black box with the same design aspect as the lo-fi prototype was upheld. It was decided to eventually put a game in the box due to the

literature research which stated that video games can improve intrinsic motivation, based on the self-determination theory. Intrinsic motivation was needed in our product to encourage users to engage with the product, with the additional goal of repeated use. Based on a different study, it was found that improving autonomy can be done within video games by giving players options and choices. One important criteria to fill was that there should be no obligatory use, captured in the design of the game containing multiple levels that do not need to be completed, or with the option to be completed multiple times. With this freedom of action, the user has no obligation to reach a certain point in the game, increasing their autonomy.



Figure 4: Archery game

The game that was created for this prototype was coded in processing. An Arduino code gathers all the data from the sensors, this data is then passed on to processing. The game has five levels, which increase with difficulty. Once all five are completed, the first level starts again. The visuals are simplistic, the background has two solid colors and the targets are 2D. Due to time constraints, it was kept this simple, however, it conveys enough for the user to understand what they are looking at. A screenshot from the game can be seen in Figure 4. There is a red circle cursor which the user can manipulate to hit the targets. When a target is hit, it disappears so the user knows that they do not have to hit it again. They proceed to the next level once all the targets are hit.

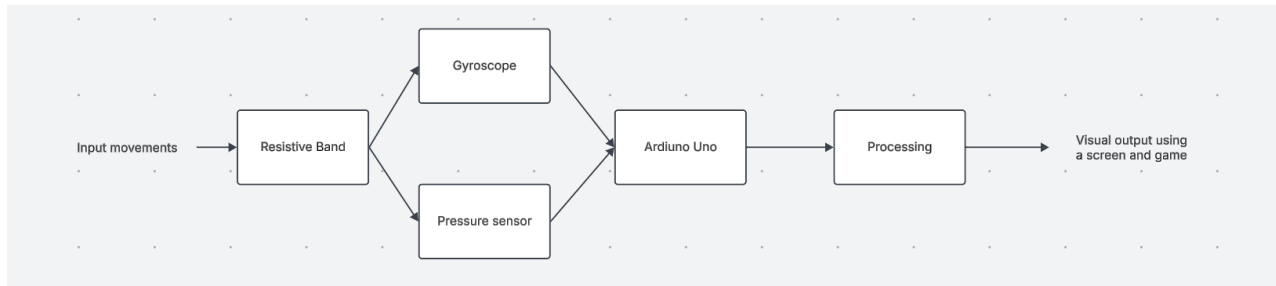


Figure 5: Architecture diagram

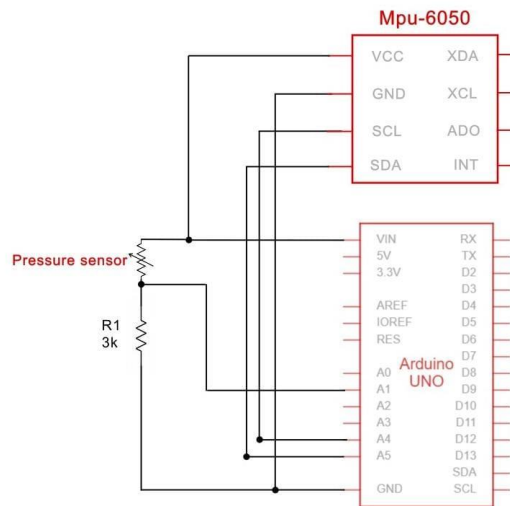


Figure 6: Circuit diagram

In Figure 5, the architecture diagram of the system can be seen. The system includes a resistance band, on which a gyroscope is mounted with a pressure sensor attached to the back of the band. These sensors attach to an Arduino UNO as can be seen in Figure 6, which shows the circuit diagram. The Arduino is connected to a laptop which displays the game that was built. The Arduino code can be found under the Appendix. This Arduino code sends information from the sensors to the processing code, in which the game was made. The game uses this information to move the red cursor.

From a technical point of view, the prototype uses two sensors as an input for the arcade game. The combination of these two sensors creates a special game controller. These sensors are a gyroscope and a pressure sensor. To clarify the placements of the sensors, a sketch has been made, this can be seen in Figure 7. The gyroscope is mounted on the side of the resistive band, using a 3D-printed holder, which can measure rotation of the band in the pitch (Y-axis) and yaw (Z-axis) through its built-in gyroscope. The roll (X-axis) could not be measured by the gyroscope, instead it was measured by the internal accelerometer. Since this measures

acceleration, not position, this direction has some offset in the prototype. Just like a bow, when pulling the band to the right and down, the arrow will point to the left and up. The pressure sensor is mounted between a 3D-printed cylinder and the resistive band. This way when the band is pulled back, the pressure sensor can sense how much pressure is being put on it. For this a special kind of pressure sensor needed to be used. It needed to be able to bend so it could equally measure the pressure from the band. The sensor was built using Velostat and copper tape. Lines of copper tape conduct electricity and Velostat acts as a resistance between them. When pressed, the value of the resistance goes down, which makes it a variable resistor. Combined with another resistor, this forms a voltage divider that allows Arduino to measure the difference in output voltage when the pressure is applied.

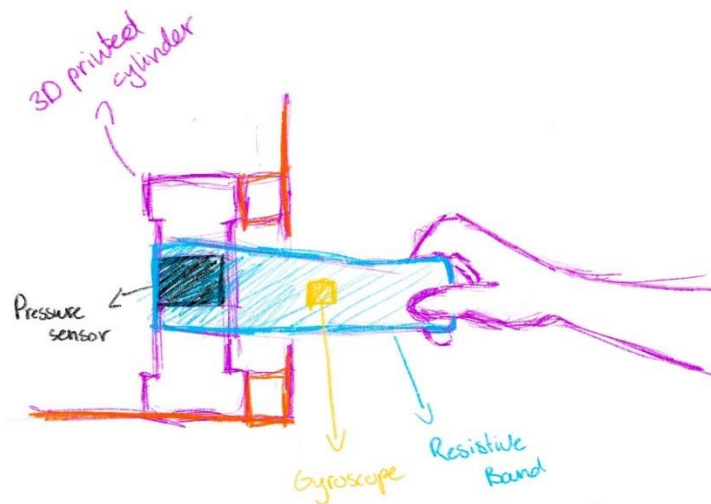


Figure 7: Sensor placement

The interaction between our Hi-Fi prototype and a user is a particular experience that depends a little on the user. First off, the user needs to notice the prototype standing somewhere in their office break room or hallway. If they have some time to spare, the user can decide whether to approach the black box and see what it does. Once they are in front of it, they will see a screen with targets and a short introduction on how to operate the resistive band. The user then will start pulling on the resistance band and see the red cursor move around the screen. This initial experimentation will show the user how to aim the cursor towards the targets. Now the user is moving the resistance band and engaging their muscles while doing so. Once the user eliminates all the targets in the first screen, they are introduced to the second level which increases the difficulty. As the levels are not infinite, once all the levels are completed the game returns to the first level. This means that the user can continue to play if they want to, though

they can also stop playing halfway through the levels. Once the user decides that they have done enough activity, they can stop playing.

5. Evaluation – Experimental Research Study

5.1 Evaluation plans

We decided that having two evaluations would be the best option for our project. One of the evaluations was completely dependent on whether people approached our product. Since there was no guarantee that anyone would approach it, there needed to be a back-up plan. This plan was an alternative evaluation in which we tested the system itself to see if it functioned correctly, without testing whether people approached the Hi-Fi prototype.

5.1.1 First experiment

The primary goal of the first evaluation experiment was to discover whether the system improved the autonomy of the user. The secondary goal was to ascertain if there was an increase from the baseline in the frequency of exercise by users when a new system was presented to them. We chose a set of dumbbells to use as the baseline device for the experiment. We wanted our system to be more accessible than gyms for office workers. As it was not realistic to additionally ask the participants to perform a gym session, dumbbells were a simplified alternative option.

The research questions that were formulated according to these goals are:

1. How much does the system improve the autonomy of the users according to the Psychological Need States in Sport Scale (PNSSS)?
2. What is the difference in usage time between the system and the baseline?

For each of the research questions, we formulated null and alternative hypotheses:

Research question 1:

H_0 : There is no difference in the results of PNSSS autonomy questionnaire between our system and the baseline or the system provides less autonomy.

H_1 : The system provides more autonomy than the baseline based on PNSSS autonomy questionnaire.

Research question 2:

H₀: There is no difference in the time users interact with the system and the baseline or the users interact with the baseline for longer.

H₁: The time users interact with the system is higher than the time they interact with the baseline.

Independent variables are our system and a set of dumbbells. Dependent variables are the time of the interaction and the perceived autonomy.

For the first experiment, we initially chose a between-subject design to decrease the required time for participation and to decrease the chance of participants guessing the purpose of the study to prevent possible subsequent changes in their behavior. However, during the experiment we realized that it was difficult to recruit participants to interact with just a set of dumbbells. Given this discovery, we found it prudent to switch to a within-subject design. Within-subject design was also appropriate for the experiment, since there was no learning effect present in the experiment, due to the independent variables being different systems.

The Hi-Fi prototype was placed in the SmartXP classroom since at the time of the experiment it was used by students who were working on their project. This provided the research group with an opportunity to recruit the students for the evaluation, easing the process of finding participants. The experiment started at approximately 2PM and continued until approximately 5PM. During this time frame, the system was turned on and participants were recruited.

Even though the designated users of this product were office workers, the first experiment used students as an easily accessible resource. This means that the results were not fully representative of the efficacy of the system, but it allowed us to arrive at a baseline result before commencing the following experiment, in which the participants we envisaged for the next step in the product development were office workers. To be able to access subjects who were employed in offices, the experiment needed to be conducted at a period in the day when most of the workers would be outside of their offices. It was not completely certain whether the system would be ready before the break on Friday 17/01/25. Hence the participation of students in the SmartXP became the most secure option to obtain data, therefore for this evaluation, the only criteria necessary for the participants to fulfill was that they needed to be students. The participants were recruited by spontaneously asking them to participate in a study. Upon agreeing to participate, they were given a consent form before the experiment started.

The participants were asked to sit down on a chair, then instructions were given to them by one of the researchers. Firstly, the researcher told them that the system was an archery game and that the controller was the resistance band in front of them. They were then instructed to pull on the band to control the cursor and attempt to aim it towards one of the targets. It was explicitly stated that the band needed to be pulled with some force as otherwise the arrows would not be shot. If the instructions were not clear enough for the participant, then the researcher was allowed to demonstrate how the system should be operated. The participant was subsequently asked to play the game for as long as they wanted. Once they announced that they were finished, they were asked to fill in the PNSSS scale. The second task given to the participants was to manipulate the dumbbells as they wanted, and for as long as they wanted, after which they were finally asked to fill in the PNSSS.

As previously mentioned, the dependent variables in this experiment were the time of the interaction and the perceived autonomy. To measure the perceived autonomy of the system or the dumbbells, the Psychological Need States in Sport Scale (Bhavsar et al., 2020) was used as a type of numerical measurement. The scale, in addition to measuring autonomy, also quantifies competence and relatedness, which are the two other pillars of the Self Determination Theory (SDT). Since we were not focused on the evaluation of these parameters for our prototype, the questions about competence and relatedness were removed, therefore the scale used for this experiment was no longer valid. In addition, the wording of some questions was changed. For example, the question that started with “In my main sport, I...”, was changed to “During the sport activity, I ...”. This change attuned the questions to the purpose of the experiment; however, it nullified the validity of the questionnaire. Nevertheless, we felt it was a necessary modification to prevent confusion, as there was a high probability that the participants would fill in the scale based on their main sport and not based on the activity they just performed. Another point worth mentioning is that the PNSSS questionnaire evaluates how autonomous users feel while performing a sports activity, while we were predominantly interested in observing to what extent the system lowered the threshold for exercising. We determined that the use of this questionnaire was the best option to obtain the data we needed, but these limitations had to be considered when analysing the results.

The time participants interacted with a system was measured using a stopwatch, starting when they received all the instructions and stopping when they said that they had finished each activity.

The first experiment had a total of ($n = 12$) participants, all of whom were students. Since the number of students participating was small, it is not possible to conduct a parametric test for analysis of the results. For this reason, the results results were analysed using Wilcoxon Signed-Rank Test.

5.1.2 Second experiment

The second experiment was conducted having office workers as participants. The goals for the second experiment were the same as for the first, but for one important modification. In this instance we would not specifically ask participants to interact with the system, as we wanted to discover whether office workers would approach the prototype without prompting. Additionally, we wanted to find out whether users would return to the system throughout the day. The research questions that were formulated according to these goals were:

1. How much does the system improve the autonomy of the users according to the Psychological Need States in Sport Scale (PNSSS)?
2. What is the difference in usage time between the system and the baseline?
3. What percentage of office workers who walk by the system will interact with it, compared to the dumbbells?
4. How many users will use the system more than one time?

For each of the research questions, we formulated null and alternative hypotheses:

Research question 1:

H_0 : There is no difference in the results of PNSSS autonomy questionnaire between our system and the baseline or the system provides less autonomy.

H_1 : The system provides more autonomy than the baseline based on PNSSS autonomy questionnaire.

Research question 2:

H_0 : There is no difference in the time users interact with the system and the baseline or the users interact with the baseline for longer.

H_1 : The time users interact with the system is higher than the time they interact with the baseline.

Research question 3:

H₀: The number of office workers who interact with our system will be the same or lower than those who interact with the dumbbells.

H₁: More office workers will interact with our system than with the dumbbells.

Research question 4:

H₀: The number of office workers who use our system more than one time is the same or lower than the dumbbells.

H₁: More office workers will use our system multiple times than engage with the dumbbells.

Independent variables were once again our system and a set of dumbbells. Dependent variables were the time of the interaction, the perceived autonomy, the number of people who interact with the system/dumbbells and the number of people who use the system/dumbbells multiple times.

For the second experiment, we used a between-subject design. The system was put in a break room for office workers in the RAM department of the University of Twente (UT). Then the dumbbells were put in another break room of the IMS department of UT. The researchers were observing the two setups from somewhere nearby where it would not be obvious that they were related to the experiment.

For this evaluation, the researchers gave no instructions or tasks to the participants. The participant walked into their break room and saw the system standing on the tables. At this juncture, it was up to the participants to decide whether they wanted to approach the device or not. When they approached the system, they were met with a screen with targets on it, and with instructions next to the product on how the resistance band should be used as a controller. If a participant chose to use the system, they were able to play the game for as long as they desired. During this period, at least two researchers were observing the participants. When a participant had finished interacting with the game, one of the researchers approached them to ask for consent to use their data. Once they agreed, we also asked them if they wanted to fill in a PNSSS questionnaire about the activity that they just performed.

The baseline that this was compared to was again the dumbbells, which were placed in a different break room. The setup was similar, as they were placed on a table. When office workers walked into their break room, they could decide whether to approach and interact with the dumbbells or not. Again, the participant was free to decide for how long they engaged with the activity. This was also observed by at least two researchers. After a participant put down the

dumbbells, one of the researchers approached to ask for their consent. If they gave consent, they were also asked to fill in a questionnaire.

The second experiment had a total of (n = 15) participants who interacted with the system and (n=0) participants who interacted with the dumbbells. All the participants were office workers. Since no one interacted with the dumbbells, it is not possible to say anything about how our system compares with the baseline. For this reason, the only analysis we can do is descriptive statistics of the results of the system.

5.2 User study

The results of the first experiment were the following:

Duration of usage System	Time
Participant 1	3m 56s
Participant 2	1m 20s
Participant 3	4m 09s
Participant 4	1m 50s
Participant 6	1m 20s
Participant 7	3m 56s
Participant 8	1m 35s
Participant 9	3m
Participant 10	3m 43s
Participant 11	2m
Participant 12	1m 30s

Table 2 The system's usage duration results of experiment 1

Duration of usage Dumbbells	Time
Participant 5	1m 03s
Participant 6	44s
Participant 7	1m 08s
Participant 8	15s
Participant 9	52s
Participant 10	11s
Participant 11	1m 42s

Participant 12	42s
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Table 3 The dumbbells' usage duration results of experiment 1

A Wilcoxon Signed-Rank Test is used to assess whether there is a significant difference between the usage of the system and the usage of the dumbbells. However, the test needs the same number of results for both conditions, therefore in the Wilcoxon Signed-Rank Test only participants 6 to 12 were used.

Results of the Wilcoxon Signed-Rank test indicated that there was a significant large difference between the usage time of the system ($Mdn = 44s$, $n = 7$) and the dumbbells ($Mdn = 120s$, $n = 7$), $W+ = 28$, $p = .008$, $r = 0.9$. Average of differences was 98.6s and SD of differences was 72,9s. Based on this we were able to reject the null hypothesis about the usage time that states that are the same.

To analyse the PNSSS two Wilcoxon Signed-Rank tests were made, one for the autonomy satisfaction and one for the autonomy frustration. First the autonomy satisfaction scores of the system and the dumbbells were compared for participants 6 to 12. Results of the Wilcoxon Signed-Rank test indicated that there was a non-significant small difference between the system's ($Mdn = 5$, $n = 7$) and the dumbbells' ($Mdn = 5.6$, $n = 7$) satisfaction scores, $Z = 0.6$, $p = .264$, $r = 0.3$.

The next step was to assess the autonomy frustration. Results of the Wilcoxon Signed-Rank test indicated that there was a non-significant medium difference between the system ($Mdn = 3$, $n = 7$) and the dumbbells ($Mdn = 3.6$, $n = 7$) frustration scores, $Z = 1.1$, $p = .853$, $r = 0.4$.

The results indicated that the null hypothesis regarding PNSSS formulated in the evaluation plan was unable to be rejected

The results of the second experiment were as follows:

Duration of usage System	Time
Participant 1	1m 30s
Participant 2	1m 10s
Participant 3	20s
Participant 4	58s
Participant 5	1m 29s
Participant 6	1m 08s
Participant 7	1m 13s

Participant 8	20s
Participant 9	40s
Participant 10	2m 00s
Participant 11	36s
Participant 12	22s
Participant 13	16s
Participant 14	1m 25s
Participant 15	1m 10s

Table 4 The system usage duration results of experiment 2

Out of 32 office workers who walked by the system, 46.9% (n = 15) interacted with it. Out of these, 26.7% (n = 4) returned to use the system again. Average usage time was $M = 58.47s$, $SD = 30.47s$

The average value of PNSSS autonomy satisfaction for the system in the second experiment was $M = 4,96$ and for the autonomy frustration, it was $M = 2,62$.

Out of 31 office workers who walked by the set of dumbbells, 0% (n = 0) interacted with them. Due to this, it was impossible to make any comparisons about perceived autonomy and usage time between the system and the dumbbells.

Additionally, some qualitative observations were made during the second experiment. Many participants demonstrated positive emotions by smiling or saying how the system was interesting, or funny. One of the office workers who interacted with the system recommended it to a different person. However, one of the participants said that the system looks sad.

Based on the usage time results of the first experiment, we can say that the system was more engaging since participants used it for longer periods of time. It is, however, important to keep in mind that the participants in the first experiment were students, not the actual intended users of the system.

The results of PNSSS of the first experiment showed no significant difference between the autonomy scores of the system and the set of dumbbells. However, as mentioned in the evaluation plan, the scale is not representative of the type of autonomy we tried to design for, and once again, the participants were not the designated users of the system. These limitations make the results non-representative of what we tried to evaluate during the experiment

Since there were no participants for the dumbbells in the second experiment, we cannot say anything meaningful about the usage time or PNSSS score results of our system. However, the fact that more people interacted with the system than with the dumbbells and that multiple participants used the system twice shows that it is more appealing to use.

Overall, results showed that users interact with our system for longer periods of time and more often compared to the set of dumbbells. However, due to multiple limitations of the conducted experiments, more evaluations need to be done to draw a definitive conclusion.

6. Discussion

In summation, we developed a project with the focus of addressing the bone density deficit in office workers, which is generally caused by their tendency to be inactive throughout a workday and continuing that trend outside of work hours. For this, we investigated autonomy, and what generates motivation in someone, so that we could design a device which would make use of these psychological mechanisms in a way which would increase an office worker's number of active minutes in a day. Through repeated testing and research, we evolved our idea and vision to the point of having a clear goal which could be worked upon and producing a device which satisfies our set requirements.

Overall, the global requirements that were categorized as a “must” requirement were correctly followed and satisfied in a way which lets us conclude that the project was a success for us. Despite some necessary changes in the requirements due to the multiple changes in the project's direction, the last version of these requirements was taken into consideration during all the stages of the development process and ended up representing the main features of the device. The system is accessible to office works since it can be easily put in a breakroom or hallway. It is this accessible due to the system being able to fit within one square meter and it being a short experience of 2-15 minutes. With the combination of the resistive band and sensors, we created a game controller that provides resistive exercises using pulling motions. Furthermore, the system is also intriguing for the users, due to its design. These are the main features that make up the final product.

If this project were to be continued, the first problem that would need to be addressed is the reliability of the system. The gyroscope and acceleration sensors both functioned poorly and experienced inaccurate input drift when working. Furthermore, the material which sensed the pressure of the resistance band was unstable and dysfunctional. The stability of the gyroscope could have been improved by using a six-axis gyroscope instead of the chosen three-axis

gyroscope. The improvement of the pressure sensor requires additional research. One of the principal limitations of the system was its inability to support more than one player, which could have been resolved with the introduction of a second controller, letting people play together in a similar fashion to a ping pong table. In addition, having a second controller could also improve the system for people playing alone, as it allows them to have more in depth controls, as well as exercising both arms at the same time. Given these changes in the design of the control scheme and the multiplayer support, more games would be made available, contributing to growing the variety of the device. We also believe that the system's dimensions could be reduced, as in its current form, the measurements for the system are 66x55x30cm, making it difficult to install non-destructively in break rooms. The dimensions of the device were a direct result of the monitor we used to display the game, which determined the width and height. We believe that by having a screen specifically designed for the system, it could have its size halved while maintaining all the same functionalities. With this change, it would be easier to install this into break rooms, which may not have the space for a larger system like a ping pong table. The goal of making the height of the device adjustable was not met due to time constraints, however it would have allowed for a more versatile set of exercises which target different muscle groups. Therefore, it would be a priority if we had more time to develop the system.

Another element which should have received more attention was the research portion of the development, due to this the research portion has some limitations that we need to consider. Our sample sizes for prototypes and interviews were insufficient to determine wholly if the results are reliable and correct. Due to the changing objectives, some of the lo-fi prototypes were conducted on people which were not part of our chosen user group of office workers. Furthermore, the baseline that was used is not sufficient, meaning that there are no significant quantitative statistics comparing our system to any others. With additional time for development, it would also have been ideal to measure the effectiveness of the device on office workers after regular use for an extended period of time, as well as the statistics regarding how many people returned to the system. With a larger budget, our system could additionally be compared to other conventional recreational systems such as ping pong or foosball tables. Lastly, the PNSSS that was used during the user studies does not measure the autonomy we intended to design for. The PNSSS measures autonomy frustration and satisfaction that users experience during a sports activity. Our product was intended to increase autonomy by lowering the threshold of starting a physical activity. The activity itself was not supposed to increase perceived autonomy. So, the results of this scale need to be interpreted cautiously. From the results of this scale, it cannot be said that the threshold of using our product was lowered.

As a result of this project, we have some key takeaways to consider in future endeavours, such as interaction technology's ability to allow for users to perform exercises which could potentially help with low bone mass problems. Our system, to some extent, addresses this problem and could be a suitable option for this topic. Another key point was the concept of autonomy and its strong impact on a person's behaviour, as shown by our interviews, literature research and project as a whole. Therefore, if we were to design another device which focuses on inciting physical activity, autonomy would be at the forefront of the design methodology. Finally, we have discovered upon developing this project that there are many unknown variables concerning bone strengthening exercises done by means of interaction technology in the context of HCI.

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8. Appendix

Appendix 2.1: Questions asked after first prototype

User 1:

1. Vertical pulling
2. Middle level pulling
3. Feet level pulling
4. Rotating
5. Middle level Pushing

User 2:

1. Feet level pulling
2. Vertical pulling
3. Pushing
4. Middle level pulling
5. Rotating

User 2 was inconsistent with his answers, moving 'feet level pulling' from 5 to 1 suddenly.

User 3:

1. Middle level pulling
2. Vertical pulling
3. Pushing
4. Feet level pulling
5. Rotate

User 4:

1. Feet level pulling
2. Vertical pulling
3. Middle level pulling
4. Pushing
5. Rotating

Observation: Users assume that pulling from the bottom is supposed to be a biceps curl

User 5:

1. Feet level pulling
2. Middle level pulling
3. Vertical pulling
4. Rotating
5. Pushing

User 6:

1. Pulling from top
2. Pulling from middle
3. Rotating
4. Pulling from bottom
5. Pushing

User 7:

1. Pulling from middle
2. Pulling from top
3. Pulling from bottom
4. Rotating
5. Pushing

User 8:

1. Pulling from bottom
2. Pulling middle
3. Pulling from top
4. Rotating
5. Pushing

Note: 1,2,3 feel quite similarly satisfying

Appendix 2.2: Context analysis interview questions

- 1) How long did/have you been going to the gym?
- 2) Why did you start going to the gym?
- 3) How does a typical gym day look like?

- 4) What does a typical day that you skip the gym look like?
- 5) Why did you drop out? ***only dropouts***
- 6) What did the process of dropping out look like? *** only dropouts***
- 7) What do you like/dislike about the gym?
- 8) What is the least fun when you're at the gym?
- 7) Did you use a stat tracker? If not, do you think that might have helped you?
- 8) Did you go alone or with friends?
- 9) Do you talk to strangers in the gym, if yes, for what reasons did you talk to someone at the gym?
- 10) Would like you to be more social in the gym? With follow-up questions to explain

Appendix 2.3: Goals of participants

The following goals were given by dropout participants, followed by how many had this goal. After this is if the goal is broad or specific, discussed by the researchers. Note 1 participant did not note any goals.

To get more fit I (broad)

Moving/doing sport II (broad)

Staying fit I (broad)

Being more healthy I (broad)

Get more muscular I (specific)

Prepare for competition I(specific)

Appendix 2.4: Results empathizing interviews

Ages:

56, 21, 20, 55, 54, 20, 19, 26, 20, 20, 21, 20, 25

Genders:

6 male, 7 female

Dropouts

Extrinsic motivations:

Get more fit I

Going with friends IV

None I

Not mentioned II

Started going because:

Convenience I

Friends II

Being physically active II

Get fitter I

Competitive training I

Did dropping out take long?

5//6 yes

Autonomy related issues:

Travelling III

Crowded gyms I

Waiting for machines II

Time scheduling issues II

Tired I

Did they enjoy going to the gym?

4/6 yes

1/6 too conflicted to answer

1/6 no

What they enjoyed:

Feeling of being active I

Seeing progress III

Challenge I

Friends I

The weightlifting itself I

Goals

To get more fit I

Moving/doing sport II

Staying fit I

Being more healthy I

Socializing with friends I

Get more muscular I

3/5: Broad

2/5: Specific

Extrinsic motivations:

Get more fit I

Going with friends IV

None I

Not mentioned II

Started going because:

Convenience I

Friends II

Being physically active II

Get fitter I

Competitive training I

Did dropping out take long?

5/6 yes

Autonomy related issues:

Travelling III

Crowded gyms I

Waiting for machines II

Time scheduling issues II

Tired I

What they enjoyed:

yes yes no yes half yes yes

Feeling of being active I

Seeing progress III

Challenge I

Friends I

The weightlifting itself I

Goals

To get more fit I

Moving/doing sport II

Staying fit I

Being more healthy I

Socializing with friends I

Get more muscular I

Gym goers

Extrinsic motivations:

pays subscription I

family and friends II

becoming fit II

Started going because:

friends I

time management I

equipment I

grow muscles I

wanting to be healthy I

relaxation I

losing weight I

Autonomy related issues:

time scheduling issues IIII

waiting times machines I

too busy gym I

weather I

Do they enjoy the gym:

yes 5/5

What they enjoyed:

freedom of choosing when to go and what to do II

challenging themselves I

relaxing I

environment I

Goals

pushing their limits I

be more healthy II

bigger ass I

weight loss I

muscle growth I

broad: 4/4

Appendix 2.5: Arduino code

```
#include "Wire.h" // This library allows you to communicate with I2C devices.
```

```
const int MPU_ADDR=0x68; // I2C address of the MPU-6050. If AD0 pin is set to HIGH, the I2C address will be 0x69.
```

```
float accelerometer_x, accelerometer_y, accelerometer_z; // variables for accelerometer raw data
```

```
float gyro_x, gyro_y, gyro_z; // variables for gyro raw data
```

```
int16_t temperature; // variables for temperature data
```

```

void setup() {

  Serial.begin(9600);

  pinMode(A1, INPUT);

  Wire.begin();

  Wire.beginTransmission(MPU_ADDR); // Begins a transmission to the I2C slave (GY-521
board)

  Wire.write(0x6B); // PWR_MGMT_1 register

  Wire.write(0); // set to zero (wakes up the MPU-6050)

  Wire.endTransmission(true);
}

void loop() {

  Wire.beginTransmission(MPU_ADDR);

  Wire.write(0x3B); // starting with register 0x3B (ACCEL_XOUT_H) [MPU-6000 and MPU-6050
Register Map and Descriptions Revision 4.2, p.40]

  Wire.endTransmission(false); // the parameter indicates that the Arduino will send a restart. As
a result, the connection is kept active.

  Wire.requestFrom(MPU_ADDR, 7*2, true); // request a total of 7*2=14 registers

  // "Wire.read()<<8 | Wire.read();" means two registers are read and stored in the same variable

  accelerometer_x = Wire.read()<<8 | Wire.read(); // reading registers: 0x3B (ACCEL_XOUT_H)
and 0x3C (ACCEL_XOUT_L)

  accelerometer_y = Wire.read()<<8 | Wire.read(); // reading registers: 0x3D (ACCEL_YOUT_H)
and 0x3E (ACCEL_YOUT_L)

  accelerometer_z = Wire.read()<<8 | Wire.read(); // reading registers: 0x3F (ACCEL_ZOUT_H)
and 0x40 (ACCEL_ZOUT_L)

  temperature = Wire.read()<<8 | Wire.read(); // reading registers: 0x41 (TEMP_OUT_H) and
0x42 (TEMP_OUT_L)

```



```

gyro_x = Wire.read()<<8 | Wire.read(); // reading registers: 0x43 (GYRO_XOUT_H) and 0x44
(GYRO_XOUT_L)

gyro_y = Wire.read()<<8 | Wire.read(); // reading registers: 0x45 (GYRO_YOUT_H) and 0x46
(GYRO_YOUT_L)

gyro_z = Wire.read()<<8 | Wire.read(); // reading registers: 0x47 (GYRO_ZOUT_H) and 0x48
(GYRO_ZOUT_L)

float aConstant = 0.0054;

float gyroConstant = 0.002;

Serial.write(128);

int gyroReading = int(gyro_y*gyroConstant);

if(gyroReading == 128) gyroReading = 129;

Serial.write(gyroReading);

int accReading = int(accelerometer_x*aConstant);

if(accReading == 128)

    accReading = 129;

Serial.write(accReading);

int markoReading = floor(analogRead(A1)/4);

if(markoReading == 128) markoReading = 129;

Serial.write(markoReading);

delay(20);

}

```

Appendix 2.6: Processing code

```
import processing.serial.*;
```

```

Serial myPort;

Levels[] levelList = new Levels[6];

Cursor cursor = new Cursor();

int levelIndex;

int yAngle, xSpeed;    // Data received from the serial port

float previousAvgY, previousAvgX;

int readingX, readingY;

float xPosition;

int yPosition;

int visibleX = 0;

int visibleY = 0;

//added

int[] markoReadings = new int [10];

int n = 0;

int p = 0;

int markoAvg;

int lastMarkoAvg;

int sum;

boolean pressed;

int readingMarko;

int timer;

void setup() {

    fullScreen();

```

```

// size(800, 800);

fill(255, 255, 255);

String portName = Serial.list()[0];

myPort = new Serial(this, portName, 9600);

for (int i = 0; i < levelList.length; i++) {

    levelList[i] = new Levels(i);

}

levelIndex = 0;

}

```

```

void draw() {

    background(0, 115, 252);

    fill(22, 106, 0);

    stroke(0);

    strokeWeight(4);

    rect(0, height/2, width, height/2);

    if (levelList[levelIndex].isLevelFinished()) {

        if (levelIndex < 4) {

            levelIndex++;

        } else {

            levelIndex = 0;

        }

    }

}

```

```
levelList[levelIndex].display();
```

```
// the bullshit
```

```
if (myPort.available() > 3) { // If data is available,
```

```
    int[] readList = new int[4];
```

```
    for (int i = 0; i < 4; i ++)
```

```
        if (myPort.available() > 0)
```

```
            readList[i] = myPort.read();
```

```
    println(readList);
```

```
    int messageStart = messageStart(readList);
```

```
    if (messageStart == 0) {
```

```
        readingY = readList[1];
```

```
        readingX = readList[2];
```

```
        readingMarko = readList[3];
```

```
    } else if (messageStart == 1) {
```

```
        readingX = readList[3];
```

```
        readingY = readList[2];
```

```
        readingMarko = readList[0];
```

```
    } else if (messageStart == 2) {
```

```
        readingX = readList[0];
```

```
        readingY = readList[3];
```

```

    readingMarko = readList[1];
} else if (messageStart == 3) {
    readingY = readList[0];
    readingX = readList[1];
    readingMarko = readList[2];
}
myPort.clear();
yAngle = readingX;
xSpeed = readingY;
if (yAngle > 128) {
    yAngle = yAngle - 255; //negatives
}
if (xSpeed > 128) {
    xSpeed = xSpeed - 255;
}
if(xSpeed < 0) xSpeed *= 1.5;
xSpeed *= 2.5;
if(xPosition > 0.5* width && xSpeed > 0) xSpeed =0;
if(xPosition < -0.5* width && xSpeed < 0) xSpeed =0;

xPosition += float(xSpeed);
xPosition *= 0.998;
previousAvgY = previousAvgY + (yAngle-previousAvgY)*0.03;
//previousAvgX = previousAvgX + (xPosition-previousAvgX)*0.03;

```

```

    //xPosition = int(previousAvgX);

    yAngle = int(previousAvgY);
}

int yPosition = int(cos(HALF_PI + radians(yAngle))*520);

float retentionRate = 0.98;

if (abs(visibleX - xPosition) > 10) {
    visibleX = constrain(int(visibleX*retentionRate + xPosition*(1-retentionRate)), -
width/2,width/2);
}

if (abs(visibleY - yAngle) > 10) {
    visibleY = int(visibleY*retentionRate + yPosition*(1-retentionRate));
}

for (int i = 0; i<markoReadings.length; i++) {
    sum += markoReadings[i];
}

println("sum" + sum);

markoAvg = sum / markoReadings.length;

sum = 0;

if (markoAvg - lastMarkoAvg >= 10 && lastMarkoAvg!=0) {
    if (!pressed) {
        pressed = true;

        timer = millis();
    }
}

```

```

    } else if (lastMarkoAvg - markoAvg >= 10) {

        pressed = false;

    }

    //if (millis() > timer + 3000) {

    // println("Pulled for 3 sec");

    //}

    markoReadings[n] = readingMarko;

    println("markoReading" + readingMarko);

    println(pressed);

    println("lastAvg" + lastMarkoAvg);

    println("avg" + markoAvg);

    n++;

    if (n >= 10) {

        n = 0;

    }

    p++;

    if (p>=75) {

        p=0;

        lastMarkoAvg = markoAvg;

    }

    cursor.update(visibleX + width/2, -yPosition + height/2, pressed, levelList, levelIndex);

    cursor.display();

}

```

```
int messageStart(int[] readList) {
    for (int i = 0; i < 4; i++)
        if (readList[i] == 128) return i;
    return 69;
}
```

```
void keyPressed() {
    if (key == 'e') {
        levelIndex++;
    }
}
```

```
class Cursor {
    float x;
    float y;
    float cursorSize;
    int timerStart;
    boolean jankFlip;
    float progressBar;
    float tensionAmount;
    int threshold;

    Cursor() {
        x = width/2;
```



```
y = height/2;  
cursorSize = 100;  
jankFlip = false;  
threshold = 80;  
}
```

```
void update(int x, int y, boolean tension, Levels[] list, int index) {  
    this.x = x;  
    this.y = y;  
  
    if(tension) {  
        cursorSize = 75;  
    }  
  
    if(tension && !jankFlip) {  
        timerStart = millis();  
        jankFlip = true;  
    } else if (!tension && jankFlip) {  
        jankFlip = false;  
    }  
  
    noStroke();  
    fill(172);  
    progressBar = map(millis() - timerStart, 0, 3000, 0, 100);
```

```

if(tension) {

    rect(x - progressBar/2, y - cursorSize - 25, progressBar, 25);

    if(millis() - timerStart >= 3000) {

        timerStart = millis();

        shot(list[index]);

    }

}

}

```

```

void display() {

    noFill();

    strokeWeight(10);

    stroke(255, 0, 0);

    ellipse(x, y, cursorSize, cursorSize);

}

```

```

void shot(Levels level) {

    level.triggerTargets(x, y);

}

```

```

}

```

```

class Levels {

    int currentLevel;

    Target[] targets = new Target[8];

```

```
int hitTargetCount;
```

```
int targetCount;
```

```
Levels(int currentLevel) {
```

```
    this.currentLevel = currentLevel;
```

```
    switch(currentLevel) {
```

```
        case 0:
```

```
            newTarget(width/2, height/2, 0);
```

```
            break;
```

```
        case 1:
```

```
            for (int i = 0; i < 3; i++) {
```

```
                newTarget(200, width/4 + 220 * i, 1);
```

```
            }
```

```
            break;
```

```
        case 2:
```

```
            newTarget(width/2, 5*height/6, 1);
```

```
            newTarget(width/2, height/6, 1);
```

```
            newTarget(width/6, height/4, 2);
```

```
            newTarget(5*width/6, height/4, 2);
```

```
            break;
```

```
        case 3:
```

```
            newTarget(width/2, height/4, 2);
```

```
            newTarget(width/4, height/4, 0);
```

```
            newTarget(3*width/4, height/4, 0);
```

```

    newTarget(width/4, 3*height/4, 0);
    newTarget(3*width/4, 3*height/4, 0);
    break;
case 4:
    newTarget(width/6, height/4, 2);
    newTarget(width/2, height/4, 2);
    newTarget(5*width/6, height/4, 2);
    newTarget(width/2, height/4, 1);
    newTarget(3*width/4, 3*height/4, 1);
    break;

}
}

```

```

void display() {
    for(int i = 0; i < targetCount; i++) {
        targets[i].display();
    }
}

```

```

void newTarget(float x, float y, int type) {
    targets[targetCount] = new Target(x, y, type);
    targetCount++;
}

```

```
}
```

```
void triggerTargets(float x, float y) {  
    for(int i = 0; i < targetCount; i++) {  
        if(targets[i].checkHit(x, y)) {  
            hitTargetCount++;  
            break;  
        }  
    }  
}  
}
```

```
boolean isLevelFinished() {  
    if(targetCount == hitTargetCount) {  
        hitTargetCount = 0;  
        for(int i = 0; i < targetCount; i++) {  
            targets[i].reset();  
        }  
        return true;  
    } else {  
        return false;  
    }  
}  
}
```

```
class Target {
```

```
int size;

float x;

float y;

int type;

int increment = 2;

boolean isHit;

Target(float x, float y, int type) {

    ellipseMode(CENTER);

    this.x = x;

    this.y = y;

    size = 200;

    this.type = type;

    isHit = false;

}
```

```
void display() {

    move(type);

    stroke(0);

    float decrement = size * 0.175;

    if(!isHit) {

        for(int i = 0; i < 5; i+=2) {

            fill(200, 0, 0);

            ellipse(x, y, size - i*decrement, size - i*decrement);
```

```

    fill(255);

    noStroke();

    ellipse(x, y, size - (i+1)*decrement, size - (i+1)*decrement);

  }

}

}

```

```

void move(int type) {

  switch(type) {

    case 1:

      noStroke();

      fill(127);

      rect(size/2, y, width - size, 5);

      x += increment;

      if(x < size/2 || x > width - size/2) {

        increment *= -1;

      }

      break;

    case 2:

      noStroke();

      fill(127);

      rect(x, size/2, 5, height - size);

      y += increment;

      if(y < size/2 || y > height-size/2) {

```

```

        increment *= -1;
    }
    break;
case 3:
    break;

}
}

boolean checkHit(float shotX, float shotY) {
    if(dist(shotX, shotY, x, y) < size/2 + 20) {
        isHit = true;
        return true;
    } else {
        return false;
    }
}

void reset() {
    isHit = false;
}
}

```